AI-POWERED UPI SCAM & FRAUD DETECTION SYSTEM

Course Code: AIM3170 (PBL-III) | **Domain:** FinTech Cybersecurity, Anomaly Detection & Graph Analytics | **Type:** Application-Based

I. Abstract (Summary)

This progress report outlines the architecture, implementation, and mid-term progress of an **AI-Powered UPI Scam and Fraud Detection System**. With the widespread adoption of the Unified Payments Interface (UPI) and real-time digital payment channels, financial cyber-fraud has rapidly evolved from isolated unauthorized debits into coordinated multi-stage attacks, including social engineering scams, nocturnal account takeover bursts, and rapid money mule layering chains. Traditional static rule engines struggle to identify these behavioral shifts without imposing high false-positive rates on legitimate users. Building upon our previous semester (PBL-II) foundation in machine learning pipelines and interactive user interfaces, this project implements a production-grade batch anomaly detection framework. In this phase of PBL-III, we established a 14-column UPI transaction schema and engineered a vectorized batch feature extraction engine that computes exactly 30 behavioral, temporal, and sequential features strictly before each transaction event, completely preventing future data leakage. Leveraging a pre-trained **Isolation Forest** model and pre-fitted **StandardScaler** with selective log transformations (np.log1p), the system scores multi-thousand transaction datasets, generating continuous anomaly scores and tiered risk classifications (High Risk, Medium Risk, Normal/Low Risk). An interactive Streamlit investigation dashboard was built featuring data quality validation, KPI overviews, multi-filter suspicious transaction ranking, interactive visualizations, deep-dive feature vector inspection, and complete CSV export capabilities. Comprehensive automated tests verified 100% compliance across 15 pipeline checks.

II. Introduction to Problem (What, Why, How)

Project Category: Application-Based System with Machine Learning & Graph Analytics

1. Problem Description (What)

Digital payment ecosystems—predominantly UPI in India—handle billions of low-latency transactions every month. However, their instant settlement and irreversibility make them primary targets for cyber-fraud. Common attack patterns include:
• **Social Engineering & Phishing:** Deceiving victims into transferring funds via fraudulent QR codes or payment links.
• **Account Takeover Bursts:** Rapid drainage of account balances to external accounts from unrecognized devices during unusual hours.
• **Mule Account Layering:** Routing illicit funds through multiple intermediary accounts in rapid succession to obstruct recovery.
In isolation, each transaction appears legitimate, but analyzing historical sequences, balance drainage ratios, and network connectivity reveals distinct anomalous footprints.

2. Motivation and Significance (Why)

• **High False Positive Burden:** Rigid rule-based systems generate excessive alerts, causing alert fatigue and friction for genuine customers.
• **Lack of Multi-Hop Visibility:** Traditional transaction-level filters evaluate transactions in silos, failing to detect organized mule syndicates.
• **Need for Prioritized Decision Support:** Fraud analysts require continuous risk rankings and explainable indicators to triage urgent cases effectively.

3. Proposed Solution & Architecture (How)

The application ingests monthly transaction batches and executes an automated validation and processing pipeline:
1. **Batch Schema Validation:** Validates all 14 required raw columns, checking for nulls, duplicate IDs, and numeric validity.
2. **Leak-Free Feature Engineering:** Chronologically computes 30 ML features capturing velocity, expanding sender/receiver statistics, and hop metrics strictly using prior data.
3. **Unsupervised Anomaly Inference:** Applies exact saved log transformations and scaling, followed by Isolation Forest scoring to assign continuous anomaly scores and risk tiers.
4. **Graph Layer & Dashboard:** Delivers a multi-section investigation interface with visual analytics, individual transaction inspection, and report downloads.

III. Work Done in Previous Semester (PBL-II Recap)

In PBL-II, our team developed an **AI Health Assistant & Disease-Specific Dietary Recommendation Platform**. That project established core competencies in training and deploying machine learning models, creating interactive Streamlit user interfaces, managing dual data persistence (MongoDB with local JSON fallback), and implementing input validation.

• Literature Survey Highlights (Major Works Studied)

• **Isolation-Based Anomaly Detection:** Studied tree-partitioning algorithms (Isolation Forest) that isolate anomalies through random splits, achieving linear computational complexity suitable for high-dimensional financial data without requiring labeled fraud datasets.
• **Mobile Money Benchmarking:** Evaluated synthetic financial logs (e.g. PaySim dataset) to study transaction step dynamics, balance continuity, and synthetic fraud injection methodologies.
• **Graph Modeling in Fraud:** Analyzed graph-based architectures where accounts represent nodes and transfers represent directed edges, showing that network degrees, path lengths, and cycle detection significantly boost detection of mule networks.
• **Temporal Behavioral Profiling:** Researched sliding-window velocity metrics, confirming that expanding historical averages, deviation z-scores, and short-term transaction counts provide strong discriminating signals for compromised accounts.

• Comparative Study Table

Approach / Paradigm	Detection Logic	Graph Awareness	Interpretability	Complexity	Suitability for UPI
Static Threshold Rules	Heuristic limits (e.g. Amount > $50k)	None (Siloed)	High (Rule Triggered)	O(1)	Low (High false alarms, easily bypassed)
Supervised Classifiers (Random Forest / XGBoost)	Supervised learning on historical labels	Low (Requires manual feature engineering)	Medium (Feature importances / SHAP)	O(N log N)	Moderate (Requires massive labeled fraud data; weak on zero-day attacks)
Isolation Forest (Our Base ML Engine)	Unsupervised subspace isolation	Moderate (Via engineered chain & hop features)	High (Continuous Decision Score)	O(N · t · log ψ)	**High (Detects zero-day anomalies without ground truth labels)**
Hybrid Behavioral + Graph Engine (Our Proposed PBL-III Target)	Unsupervised ML + Directed Graph Topology	**High (Direct multi-hop cycle & path tracing)**	**High (Risk Tier + Feature Vector + Graph Evidence)**	O(N log N + V + E)	**Optimal (Combines individual account profiling with network syndicate detection)**

• Objectives Set in PBL-II & Planned Methodology

1. Transition from supervised classification to unsupervised behavioral anomaly detection in financial streams.
2. Build scalable batch pipelines that process multi-thousand transactions without temporal data leakage.
3. Design an investigative dashboard tailored for operational fraud triage and audit reporting.

IV. Progress Made Till Now (PBL-III Work)

• Implementation Steps Completed So Far

1. **Schema Standardization & Input Validation:** Established the 14-column raw format (transaction_id, sender_id, receiver_id, amount, transaction_type, transaction_step, sender_old_balance, sender_new_balance, receiver_old_balance, receiver_new_balance, transaction_time, device_id, device_change_flag, new_device_flag). Built validation checks in feature_engineering.py verifying column presence, unique IDs, non-negative amounts, and datetime formats.
2. **Leak-Free Feature Engineering (30 Features):** Implemented a chronological computation engine ensuring expanding statistics (historical counts, expanding means, sample variances, z-scores) and rolling windows (1h, 6h, 24h via np.searchsorted) only aggregate prior events ($t_{prior} < t_{current}$).
3. **Exact Preprocessing & Inference:** Applied np.log1p strictly to the 11 features specified in preprocessing_config.pkl, normalized data via scaler.transform() from scaler.pkl, and generated predictions and decision scores via isolation_forest.pkl.
4. **Interactive Streamlit Dashboard:** Developed fraud_detection_app/app.py with 6 dedicated sections: Upload, Overview KPIs, Ranked Suspicious Table, Visual Analytics, 30-Feature Deep-Dive Inspector, and Scored/Anomalies CSV Downloads.

• Tools and Technologies Used

Category	Technology / Tool	Role in Project
Core Backend	Python 3.10+ / NumPy & Pandas	Vectorized batch feature engineering, statistical expanding windows, and rolling velocity counters.
Web Application	Streamlit	Batch file upload, real-time KPI metrics, interactive filtering, charts, and report exports.
Machine Learning	Scikit-Learn (Isolation Forest)	Pre-trained tree-based anomaly isolation and continuous decision scoring.
Graph Analytics	NetworkX (In Progress)	Directed account-to-account graph modeling and multi-hop layering cycle detection.
Serialization	Pickle	Dynamic loading of frozen model artifacts, scaler parameters, and feature configurations.

• Partial Results, Prototypes & Automated Verification

An automated test suite (test_batch_app.py) was developed and executed on a 500-transaction benchmark dataset (sample_transactions.csv), achieving a **100% pass rate across all 15 operational requirements**:
✓ **Artifact Integrity:** Successfully loaded all 4 pickle files without modification.
✓ **Feature Dimensionality:** Produced exactly 30 features strictly matching feature_config.pkl.
✓ **Immutability Verification:** Proved that scaler.mean_, scaler.scale_, and model.offset_ were never refitted or retrained.
✓ **Inference Validation:** Scored all 500 transactions into High Risk (score < -0.05), Medium Risk (-0.05 to 0.0), and Normal (score ≥ 0.0).
✓ **Export Functionality:** Generated complete enriched CSVs and filtered anomalies-only exports successfully.

• Challenges Faced and Addressed

• **Preventing Data Leakage:** Sorted transactions chronologically prior to computation so that counts and variances only aggregate historical transactions.
• **Distribution Skewness:** Applied np.log1p to 11 highly skewed features (amounts, balances, stats) to preserve linear scaling fidelity.
• **Computational Efficiency:** Replaced slow iterative loops with vectorized groupby operations and binary search (np.searchsorted) for rolling time windows.

V. Work in Progress / Pending Work

• Tasks Being Worked On

1. **NetworkX Directed Graph Engine:** Constructing directed money-flow graphs where accounts form nodes and transactions form weighted edges with timestamp metadata.
2. **Scam-Chain & Mule Ring Detection:** Developing topological algorithms to detect rapid multi-hop chains (A → B → C → D) and fund aggregation/dispersal hubs.
3. **Signal Explainability Module:** Creating human-readable risk attribution badges highlighting contributing factors (e.g. nocturnal transfer, new device, extreme balance ratio).

• Next Milestones

Milestone	Target Deliverable	Expected Output
Milestone 1	NetworkX Graph Layer	Account node creation, degree centralities, and mule path detection.
Milestone 2	Multi-Signal Risk Engine	Combined scoring synthesizing behavioral anomaly score with graph topology risk.
Milestone 3	Interactive Network Visualizer	Interactive node-link UI visualizing suspicious money flow paths.
Milestone 4	Database & Case Management	Persistent storage for batch audit history and flagged accounts.
Milestone 5	Evaluation & Final Report	Precision/recall benchmark evaluations and final project documentation.

VI. Bibliography / References

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